Project Team Charter

MIME 498: Capstone 2

Signature Cover Page

ME- 044: Musicalbot Encore

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: Kimberlee Britton Signature:

Written Statement:

I agrees to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team.

Printed Name: John Hansen Signature:

Written Statement:

I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team

Printed Name: Tyler Kosa Signature:

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Printed Name: Cameron Lewis Signature:

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Printed Name: Derek Martin Signature:

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Printed Name: Tim McDonald Signature:

Written Statement: I agree to do an equal amount of work in the team. I understand that my grade will reflect my effort in the team

Printed Name: Kyle Wern Signature:

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Team Purpose

As a team, we share a goal of bridging our passions for engineering, community outreach, and music. By creating a unique and highly interactive autonomously playing MusicalBot, we plan to engage K-12 students attending the 2026 Engineering Expo and inspire them to pursue a career in engineering. Through combining our skills in music creation, coding, and electronics, we aspire to create a functional prototype and memorable experience that students and their families will recall long after we graduate.

Our primary stakeholder for this project is Dr. Oman. Throughout this term, the course will challenge us to conceptualize and provide a final proof of concept we can manufacture in the last term of our undergraduate degrees. In addition to brainstorming, vetting, and finalizing a robust design for the MusicalBot, Dr. Oman expects us to document our project well and demonstrate the feasibility of our design. Additionally, to accomplish these goals, we recognize that we must communicate effectively and professionally with our stakeholders as well as with each other to meet deadlines and invent something truly remarkable.

Team Goals

Our primary goal is to finalize the design for a unique and interactive autonomous instrument that we can display at Expo. This device will not only produce music, but it will do so with an aesthetically pleasing appearance and diverse sound. Additionally, our plan to build the device from at least 50% recycled materials will reflect our commitment to sustainable and economically-conscious design. With these key factors in mind, our major goal will push us to demonstrate our resourcefulness and challenge us to think creatively with repurposed materials.

In addition to our main tasks, our “stretch goals” center around compatibility and open access. Rather than simply developing a machine that plays music alone, we strive to have it play alongside the existing GuitarBot and remain simplistic enough in design that students can adapt our findings to their own projects. These steps may include using an open-source coding language, adapting materials from basic electronics kits, and providing clear technical documentation that gives the information they need to implement our design process.

Because of our ambitious goals, our team is dedicated to upholding a high standard of performance and synchronization. We will use our time this term to strive for excellence in our design and earn As in the course. Along with collaborating on deliverables, we plan to remain accountable on our timecards and refer to this living document to recall our responsibilities.

Team Member Personalities and Roles

Our team's diverse High5 Strengths survey results provided us with insight into the strengths we bring to the table, which have been summarized in the list below.

- Kimberlee Britton: Catalyst, Empathizer, Timer Keeper, Deliverer, Analyst
- John (Jack) Hansen: Empathizer, Winner, Storyteller, Commander, Analyst
- Tyler Kosa: Believer, Problem Solver, Peace Keeper, Philomath, Coach
- Cameron Lewis: Thinker, Deliverer, Believer, Focus Expert, Catalyst
- Derek Martin: Focus Expert, Believer, Self-Believer, Philomath, Optimist
- Tim McDonald: Pacer Keeper, Phiomath, Problem Solver, Empathizer, Analyst
- Kyle Wern: Problem Solver, Believer, Philomath, Analyst, Winner

Despite these data points, we felt at odds with some of our results and uncomfortable assigning roles without a deeper discussion. For this reason, we agreed to assign these roles by preference and be open to change. For now, our assumed positions and their responsibilities are summarized in the table below (see Table 1).

Role Title	**Person(s)**	**Description**
Project Manager	Kimberlee Britton Kyle Wern	Organize meetings with the team and the sponsor by providing agendas and taking notes Check that assignments and responsibilities are distributed equally across all team members
Logistics Manager	Jack Hansen	Make sure that all assignments are turned in on time and in accordance with the rubric
Financial Manager	Kyle Wern	Manage the finances and the ordering of materials as needed
Test Engineer	Cameron Lewis Jack Hansen Tim McDonald Tyler Kosa	Test the product at multiple stages
Manufacturing Engineer	Cameron Lewis Kyle Wern Kimberlee Britton	Create parts/ensure parts are designed to tolerance

	Tim McDonald Derek Martin Tyler Kosa Jack Hansen	
CAD/FEA Engineer	Tim McDonald Kyle Wern Tyler Kosa Cameron Lewis	Manage CAD modeling and dividing up the parts Manage Finite Element Analysis processing
Computer Engineer	Derek Martin Kimberlee Britton	Manage Python operations and MIDI simulations

Table 1: Team member assignments

Kimberlee Britton, one of our team's Project Managers, brings strong organizational and communication skills to the team by coordinating meetings with both the team and sponsor, preparing agendas, and documenting progress. With her attention to detail and ability to keep schedules on track, she ensures the entire team stays aligned with deadlines and project expectations, while also checking that assignments and responsibilities are distributed equally across all team members. She also serves as a Manufacturing Engineer, where she helps create parts and verify that designs are made to tolerance, and as a Computer Engineer, where she supports Python operations and MIDI simulations essential to our design process.

As a Computer Engineer, Derek Martin manages Python operations and MIDI simulations that support the team's design and testing process. His technical experience helps maintain the digital systems needed to evaluate concepts and support development efficiently. In addition, as a Manufacturing Engineer, he helps create parts and guarantees that designs are made to tolerance.

Tyler Kosa serves as the test engineer, manufacturing engineer, and CAD engineer. His skills in analyzing mechanical components and CAD help the team create 3D printed parts and ensure our design's dimensions fall within tolerance.

With a strong mind for organization and deadlines, Jack Hansen, our Logistics Manager, verifies that assignments are submitted on time and in accordance with course rubrics. He monitors ongoing task completion, tracks required deliverables, and supports communication between members to maintain overall project flow.

Kyle Wern serves as the group's second Project Manager, Financial Manager, Manufacturing Engineer, and CAD/FEA Engineer. His enthusiasm for productive workflow; experience with budgeting, purchasing, and material sourcing; and interest in both manufacturing and CAD make him well-suited for these positions. His practical problem-solving skills and comfort with numbers allow him to handle cost decisions, evaluate material tradeoffs, and ensure that orders are placed efficiently. His skills in visualization, tolerance adherence, and design for manufacturing are also unmatched, making him an effective Manufacturing Engineer for our team.

As both the Test Engineer and a Manufacturing Engineer for the team, Cameron Lewis bridges the gap between design and production. He helps fabricate components, , assure that parts are designed to tolerance, and tests the product at multiple stages. He also serves as a CAD/FEA Engineer, where he helps manage CAD modeling, part division, and uses Finite Element Analysis to test parts' bending strength throughout the design process.

Tim McDonald serves as a Test Engineer, Manufacturing Engineer, and CAD/FEA Engineer for the team. In these roles, he tests the product at multiple stages, helps create parts and determines that they are designed to tolerance, and supports CAD modeling by helping divide up the parts across the team. He also contributes to Finite Element Analysis processing, helping strengthen the team's ability to evaluate and refine the design throughout development.

Ground Rules

To ensure we meet our deadlines on time, we have established regular meetings and communication channels. We hold meetings on Tuesdays from 4:00 – 6:00 pm and Fridays from

1:00 – 3:00 pm in a teammate's garage. The team primarily uses the Tuesday meetings to work ahead on written deliverables due each Friday and Sunday, and to complete construction after each deliverable is at least 75% complete. Fridays' meetings are similarly used for construction, as well as meeting with the team's sponsor (Dr. Oman) and the rest of the Spring 2026 Capstone teaching team every other week for a 30-minute Zoom meeting. Throughout the week and over the weekend, each member of the team has also agreed to commit at least 2 hours outside of scheduled meeting hours toward construction and testing to make sure the project progresses at an optimal rate. Because of the hands-on nature of the meetings this term, the team prioritizes in-person meetings, and team members who cannot attend meetings in person due to travel or illness are expected to attend over Zoom or Discord to continue working with other team members on written deliverables.

Between our meetings, we communicate with one another primarily in our Discord channel to draft emails, ask clarifying questions, and send updates regarding our workflow. For more formal communication with our sponsor, we use Outlook and CC members on each email. In addition, we check both our emails and the Discord channel at least once a day to certify all parties (including our mentor/sponsor) are on the same page. As an example, for major design decisions or scope changes, prioritizing in-person check-ins with Dr. Oman before proceeding will ensure that our design remains aligned with sponsor expectations and reduces the risk of late-stage redesign.

In order to foster a healthy work environment, we have agreed to adhere to a few norms centered on mutual respect. In discussions, we treat everyone as we would like to be treated, acknowledge that every idea has potential, and agree that mistakes are a natural part of the engineering process. During discussions, we make sure each member can speak without interruption, and we listen actively to facilitate productive conversations. Along with respect, we also firmly believe in honesty. When dissenting views arise, we address them through in-person discussions where we focus on critiquing the idea rather than the person. By cultivating an environment where it is safe to disagree, we ensure that our final decisions are thoroughly evaluated and that every team member feels heard and respected.

To make sure the project stays on track, we apply several accountability measures centered on transparency and time management. During weekly meetings, we review progress and identify blockers early. If a task is taking longer than expected, it is reassigned or broken into smaller components. This approach allows for steady progress and prevents inefficient use of team resources. We utilize timecards weekly to track our efforts,

and spend the first 15 minutes of every team meeting reviewing each member's weekly progress. We expect a high level of commitment, meaning everyone attends all meetings unless a valid excuse arises, in which case they notify the group as early as possible. Most importantly, we rely on total honesty regarding timecard entries to maintain trust within the team and ensure that our reported data accurately reflects our collective labor.

Finally, we use a structured plan to manage the workload and maximize our efficiency. The Project Manager is be responsible for identifying any tasks that cannot be completed on time and posting them for other members to pick up in our Discord channel thread "assignment links." To make sure no one's time is wasted, keep our meetings focused and productive by adhering to agendas and sticking to our project timelines. We also strive to distribute work in a way that aligns with each member's individual strengths and interests, making sure that everyone contributes in a way that is both effective for the project and personally engaging.

Potential Barriers and Coping Strategies

Although we are a cohesive team, we anticipate that some personal matters may affect the team's performance. As graduating seniors, our schedules are impacted by the day as we search for jobs and otherwise organize our post-graduation life. In order to reach our deadlines on time, we agree to communicate openly and honestly about our schedule limitations. For example, if a member cannot complete a task by a certain time due to personal events, that the member informs the team early so their work can be redistributed to other members. We also expect that this practice is used honestly and with the other team members in mind to avoid overwork. These expectations reflect some of our past project experiences, in which some members struggled with different teammates who did not use effective communication or follow through with their actions. In order to avoid these past cycles, we cultivate an environment where every member feels comfortable speaking openly, and we accomplish this goal by reinforcing the idea that every suggestion has value and must be met with respect. When we notice a drop in participation or a shift toward rude communication, we handle it through the in-person discussions outlined in our norms. These steps make certain that we act with integrity and address the behavior respectfully before it affects our collective progress.

Regarding our relationship with our sponsor, we anticipate that challenges like shifting schedules or differing perspectives on the project scope will arise. Since Dr. Oman has other academic and professional commitments, we remain flexible and diligent about checking in. To mitigate issues over misaligned timelines or the complexity of our group dynamics as we divide tasks, we communicate our schedules and project

planning with complete transparency. By reaching out as soon as a potential conflict or a technical disagreement occurs, we ensure that decisions are made collaboratively and that our progress remains unaffected by avoidable misunderstandings.

Prototyping Strategies

As we approach physical realization, we will test the feasibility of the design by relying on a balance of digital validation and team consensus. We plan to avoid scope creep by looking to our HoQ for reference and checking in with our sponsor. After creating our prototype in CAD, we will utilize FEA to simulate mechanical stresses and confirm that our design choices are sensible (e.g., that it can withstand multiple keys being pressed simultaneously). From here, we will meet as a team to review our final design, check with our sponsor that our design fits all criteria required, and order our materials. During this time, we will continue using digital wiring diagrams to plan exactly what components we need to order, and how exactly each of these components will interact.

In order to avoid material procurement difficulties, we plan to order critical parts as early as possible and lean on reliable suppliers. By using sites like McMaster-Carr and Amazon Prime, we can minimize downtime. Throughout this stage, we will keep Dr. Oman updated via email or in class to ensure she remains informed of our technical progress and any issues we encounter obtaining materials. Additionally, we have scheduled bi-weekly meetings with Dr. Oman to ensure that we are on schedule and aligned with project goals.

If our prototype underperforms or fails, we will react by analysing its failures systematically. We intend to quantify any underperformance immediately, brainstorming solutions as a group to decide if the issue can be solved through iteration or if it indicates a fundamental design flaw. While we will "cross that bridge when we come to it" in terms of specific fixes, we are setting ourselves up now by ensuring all qualitative observations are recorded in our shared Google Drive. This documentation strategy will ensure that the knowledge we gain during assembly is preserved, allowing us to execute the most promising solutions quickly and keep the project moving forward. We will keep Dr. Oman informed about our prototyping process through the aforementioned bi-weekly meetings that have been scheduled, and will report to her about the current state of our prototyping efforts.

Our final prototyping strategy involves wiring a full white octave and black octave for testing with a sample MIDI file (e.g., the Pirates of the Caribbean theme). This wiring process will be completed by the end of the first week of the term. During this time, the team will verify

that each element required for the assembly has arrived or is in transit. This step will allow the team to complete Testing Procedures 1, 4, and 8, as outlined in our Design-based Testing Procedures document. From the results of these tests, the team can apply the lessons learned to lay the foundation for Testing Procedures 5, 6, and 7, which assess note accuracy, mechanical noise, and playable range. Once the team completes the aforementioned testing procedures for the two octave assemblies, we will move toward adapting what we have learned to create a fully integrated 61-key system. After creating the full prototype, the team will use Testing Procedures 2 and 9 to confirm playback reliability and that the device allows users to select multiple songs that play across the instrument fully. At this time, the team will ensure the exterior remains transparent to adhere to visibility requirements using Testing 3. When these testing procedures are complete, the team will complete Testing Procedures 10 and 11 to verify that the device can be set up in 10 minutes and transported by 2 team members. Lastly, the team will use Testing Procedures 12 and 13 to check that the user interface and poster are easy for a general audience to understand. With these procedures modeled after our HoQ in mind and regular check-ins with our sponsor, the team can move forward with creating an engaging and robust design capable of experiencing repeated use at the Spring 2026 Engineering Expo.